Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester III)

Data structure

Practical – 2

Linked List Manipulation

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2. Linked List Manipulation:

Write a program to:

a. Create a singly linked list.

Input

```
class Node:
    def __init__(self,data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(15)
node3 = Node(7)
node4 = Node(10)
node5 = Node(9)

node1.next = node2
node2.next = node3
node3.next = node4
node4.next = node5
#node1.next = None

def traver(head):
    curr = head
    while curr:
        print(curr.data,end="=>")
        curr = curr.next
    print("Null")

traver(node1)
print("Priyanshi Gupta S012")
```

Output

```
11=>15=>7=>10=>9=>Null
Priyanshi Gupta S012
```

b. Insert a node at the beginning, end, and at a given position in a linked list.

Input

```
class Node:
    def __init__(self,data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(15)
node3 = Node(7)
node4 = Node(10)
node5 = Node(9)

node1.next = node2
node2.next = node3
node3.next = node4
node4.next = node5
#node1.next = None
global head
head = node1

def traver(head):
    curr = head
    while curr:
        print(curr.data,end="=>")
        curr = curr.next
    print("Null")

#insert at begining
def insert_beg(data):
    global head
    new = Node(data)
    new.next = node1
    head = new

#insert at end
def insert_end(data):
    global head
    new = Node(data)
```

```
    if head is None:
        head=new

    else:
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new

#insert at specific position
def ins_pos(pos,data):
    global head
    new = Node(data)

    temp = head
    for i in range(pos-2):
        temp = temp.next
    new.next = temp.next
    temp.next = new

insert_beg(10)
insert_end(25)
ins_pos(2,17)
traver(head)
print("Priyanshi Gupta S012")
```

Output

begining

```
10=>11=>15=>7=>10=>9=>Null
Priyanshi Gupta S012
```

end

```
10=>11=>15=>7=>10=>9=>25=>Null
Priyanshi Gupta S012
```

Specific position

```
10=>17=>11=>15=>7=>10=>9=>25=>Null
Priyanshi Gupta S012
```

c. Delete a node from a given position in a linked list.

Input

```python
class Node:
    def __init__(self,data):
        self.data = data
        self.next = None

node1 = Node(11)
node2 = Node(15)
node3 = Node(7)
node4 = Node(10)
node5 = Node(9)

node1.next = node2
node2.next = node3
node3.next = node4
node4.next = node5
#node1.next = None
global head
head = node1

def traver(head):
    curr = head
    while curr:
        print(curr.data,end="=>")
        curr = curr.next
    print("Null")

#insert at begining
def insert_beg(data):
    global head
    new = Node(data)
    new.next = node1
    head = new

#insert at end
def insert_end(data):
    global head
    new = Node(data)
```

```
    if head is None:
        head=new

    else:
        temp = head
        while temp.next:
            temp = temp.next
        temp.next = new

#insert at specific position
def ins_pos(pos,data):
    global head
    new = Node(data)

    temp = head
    for i in range(pos-2):
        temp = temp.next
    new.next = temp.next
    temp.next = new

#delete at specific poisition
def dele_pos(pos):
    global head

    temp = head
    for i in range(pos-2):
        temp = temp.next
    temp.next = temp.next.next
insert_beg(10)
insert_end(25)
ins_pos(2,17)
dele_pos(4)
traver(head)
print("Priyanshi Gupta S012")
```

Output

```
10=>11=>15=>7=>10=>9=>Null
Priyanshi Gupta S012
```

Delete the 15 value

```
10=>17=>11=>7=>10=>9=>25=>Null
Priyanshi Gupta S012
```